# Zhang Xinlei

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## EDUCATION

**Nanyang Technological University, Singapore** Aug 2025 – Jun 2026

*Master of Science in Biomedical Engineering*

Selected coursework: In Vitro Diagnostics and Bio-nanotechnology; Polymeric Materials for Biomedical Applications; Drug Delivery and Nanomedicine; Tissue Engineering and Regenerative Medicine.

**Suzhou University of Science and Technology, China** Sep 2021 – Jun 2025

*Bachelor of Engineering in Environmental Science*

## RESEARCH EXPERIENCE

**Research Assistant | Nanyang Technological University, Singapore**

Sep 2026 – Present

*Lectin-Mediated Magnetic Bacterial Enrichment Toward Rapid Antimicrobial Susceptibility Testing*

- Investigating Fc-MBL-functionalized magnetic beads for the capture and enrichment of Escherichia coli and Staphylococcus aureus at low bacterial concentrations.
- Assessing bacterial recovery and capture efficiency through magnetic separation and culture-based colony-forming unit (CFU) enumeration.
- Comparing capture performance between Gram-positive and Gram-negative bacterial models to investigate species-dependent differences and guide optimization of the bacterial enrichment workflow.

**Master's Research Project | Nanyang Technological University, Singapore**

Sep 2025 – Aug 2026

*Functional Gelatin–Chitosan Films for Food Preservation*

- Prepared biodegradable gelatin–chitosan composite films and investigated tannic acid–zinc modifications for functional packaging applications.
- Evaluated wettability, FTIR spectra, water-vapor barrier properties, antioxidant activity (DPPH/ABTS), and antibacterial performance.
- Compared formulations through fresh-cut apple preservation experiments, including weight loss and browning assessments.

**Undergraduate Research | Suzhou University of Science and Technology, China**

Mar 2023 – Jun 2025

*Isolation and Functional Characterization of Plant Growth-Promoting Rhizobacteria (PGPR)*

- Isolated and screened plant growth-promoting rhizobacteria associated with rice lines tolerant to low nitrogen and phosphorus, using culture-based techniques and 16S rRNA-based identification.
- Investigated *Bacillus altitudinis* Hxx04 and its plant-growth-promoting potential; the study reported enhanced rice seedling growth and nitrogen/phosphorus uptake in pot experiments.

**Undergraduate Research | Suzhou University of Science and Technology, China**

Sep 2023 – Jun 2025

- *Zebrafish Neurotoxicity under Micro-/Nanoplastic and Silver Nanoparticle Co-exposure*

Conducted zebrafish exposure experiments starting at 6 hours post-fertilization (hpf) for a study of combined polystyrene micro-/nanosphere and silver nanoparticle toxicity.

- Performed larval locomotion assays using EthoVision XT (distance and velocity tracking) and generated behavioral data for downstream team analysis.

**Undergraduate Research | Suzhou University of Science and Technology, China**

Sep 2024 – Jun 2025

*Isolation and Characterization of Sethoxydim-Degrading Bacteria*

- Isolated and characterized four bacterial strains from contaminated soil using enrichment culture, biochemical characterization, and 16S rRNA gene sequencing.
- Identified *Delftia sp.* Z4 as a promising degrader, with 65.6% sethoxydim degradation quantified by HPLC.

## PUBLICATIONS

**Xinlei Zhang**, Yuxin Fan, Xintong Jiang, Yuxi Yang, Hongyan Shen, Shan Zhuang, Jianheng Wang, Xinyu Liu, Jing Bai, Yanjie Qi, Dongfei Han. Isolation, identification, and application of plant growth-promoting rhizobacteria from backcross inbred rice populations with tolerance to low nitrogen and low phosphorus. *Acta Microbiologica Sinica*. 2026, 66(4): 1819–1838. DOI: 10.13343/j.cnki.wsxb.20250633

Jie Song, Qian Pu, Chen Chen, Xingcheng Liu, **Xinlei Zhang**, Zejun Wang, Jin Yan, Xuedong Wang, Huili Wang, Qiuhui Qian. Neurological Outcomes of Joint Exposure to Polystyrene Micro/Nanospheres and Silver Nanoparticles in Zebrafish. *Environmental Health Perspectives*. 2025, 133(5): 057007. DOI: 10.1289/EHP14873

Research contribution: Conducted zebrafish exposure experiments and larval locomotion assays using EthoVision XT.

**Xinlei Zhang**, Wenru Liu. Progress on nitrogen removal driving mechanism and influencing factors of Feammox. *Industrial Water Treatment*. 2025, 45(5): 1–11. DOI: 10.19965/j.cnki.iwt.2024-0526

## PATENT

**[First inventor]** Formaldehyde Catalytic Manganese-Oxygen Composite Nano Filter Material and Its Preparation Method and Application. CNIPA patent application, application no. 2024104079088.

## TECHNICAL SKILLS

**Microbiology & bacterial assays:** Bacterial culture and isolation; $OD_{600}$ measurements; serial dilution; CFU enumeration; magnetic bead-based capture; 16S rRNA-based identification.

**Biomaterials & characterization:** Gelatin–chitosan film preparation; tannic acid–zinc modification; antibacterial and antioxidant (DPPH/ABTS) assays; contact-angle, FTIR, and water-vapor barrier analysis.

**Analytical & data tools:** HPLC-based quantification; SEM, XRD, XPS and BET characterization experience; GraphPad Prism, Origin, R/RStudio, Python/Jupyter Notebook, MATLAB.

**Languages:** Mandarin (native); English.

## HONORS & AWARDS

- University First-Class Scholarship (rank 1/81), 2023–2024 academic year
- University Third-Class Scholarship (rank 19/81), 2022–2023 academic year
- National-Level College Student Innovation Training Project (2025)
- University-Level Outstanding Project Award, National College Student Energy Conservation and Emission Reduction Competition (2024)

## REFERENCES

**Prof. Hongwei Duan** | hduan@ntu.edu.sg | MSc research supervisor
Nanyang Technological University

**Prof. Dongfei Han** | dongfeihan@usts.edu.cn | Undergraduate supervisor
Suzhou University of Science and Technology